Handwritten Digit Recognition Using Deep Learning

1. Introduction. Recognising handwritten digits is a long standing problem in computer vision with applications in postal automation and cheque processing. This project builds a classifier for the MNIST dataset and evaluates how well it generalises to unseen samples collected by the authors. The work was carried out as a final year undergraduate project.

2. Methodology. We implemented a CNN in Python using TensorFlow and Keras. Images were normalised and reshaped before training. We also trained an SVM and a random forest as comparison models using scikit-learn.

3. Results. The CNN achieved an accuracy of 98.2 percent on the test split. We report precision and recall per class and include a confusion matrix.

4. Limitations. A limitation of this work is that no cross validation was performed and the model was evaluated on a single train test split. Future work should address class imbalance in the authors own samples.